

AI-powered platforms in STEM education, insights from UTAUT model: PLS-SEM and artificial neural networks hybrid analysis

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ABSTRACT

This investigation examines the effect of AI-powered technologies on STEM cognition by means of the unified theory of acceptance and use of technology (UTAUT) model. A sample of 160 respondents from diverse academic backgrounds was analysed by partial least squares structural equation modelling (PLS-SEM) to test the UTAUT model. Furthermore, artificial neural networks with a multilayer perceptron (MLP) architecture were employed to validate the outcomes and forecast the relationships amongst the UTAUT constructs. The examination focused on the core UTAUT constructs: performance expectancy, EE, SI, facilitating conditions, and their influence on the behavioural intention and actual use of AI tools in STEM education. The outcomes indicated that performance expectancy and facilitating conditions are significant predictors of behavioural intention (BI), signifying the perceived usefulness of AI tools and the existence of supportive didactic settings are critical for their adoption. Nevertheless, the conjectured moderating effects of user gender and institutional role were not strongly supported. However, the hypothesized moderating effects of user gender and institutional role on these relationships were not strongly supported, inferring a more universal applicability of AI technologies across demographic sets. These outcomes highlight the necessity of integrating AI tools into educational practices in a way that is collectively available and operative, nurturing wide-ranging educational environment that aids all users irrespective of demographic variances.

1. Introduction

The integration of artificial intelligence (AI) into education is reshaping teaching and learning paradigms globally, with particularly profound implications for science, technology, engineering, and mathematics (STEM) disciplines. AI-powered tools—such as adaptive learning systems, virtual laboratories, and intelligent content generation—are increasingly recognised for their ability to enhance computational thinking, engagement, and performance [1–3]. For example, AI-driven platforms like DreamBox, Labster, and MATLAB simulations have been shown to improve learner comprehension and retention by tailoring content, providing immersive experiences, and facilitating complex experimentation in engineering and science contexts (see Table 1).

Despite these advancements, the adoption and sustained use of AI in STEM education is influenced by a range of factors. Existing research applying the Unified Theory of Acceptance and Use of Technology (UTAUT) highlights constructs such as performance expectancy, effort expectancy, social influence, and facilitating conditions as critical

predictors of behavioural intention [18,19]. However, empirical studies examining how demographic variables—particularly gender and institutional role—moderate these relationships in STEM contexts remain scarce. Addressing this gap is important, as demographic and role-based differences could shape the pathways through which AI technologies are accepted and integrated into educational practice [4,20].

1.1. Purpose of the study

This study empirically investigates the determinants of behavioural intention and actual use of AI-powered tools in STEM education, focusing on the moderating effects of gender and institutional role within the UTAUT framework. By combining partial least squares structural equation modelling (PLS-SEM) with artificial neural network (ANN) analysis, it offers a robust examination of both linear and non-linear relationships.

Despite extensive application of the UTAUT model in educational technology research, few studies have empirically examined how gender and institutional role moderate AI adoption in STEM education within

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the South African context. Existing evidence remains fragmented and often drawn from non-STEM or non-African settings [1,5,13]. This study directly addresses this gap by combining PLS-SEM and ANN analyses, offering a dual perspective that captures both linear and non-linear effects. By focusing on South African higher education, the research advances current debates on equitable AI adoption where demographic divides may be narrowing [2,19,21].

While gender and institutional role have been studied as moderators in technology adoption elsewhere, the South African STEM higher education environment presents a distinct setting where infrastructural inequities, digital literacy divides, and institutional hierarchies operate differently from well-resourced international contexts. Thus, the contribution is not merely incremental: it tests whether previously observed patterns hold under conditions of systemic constraint and cultural diversity, providing evidence that is currently absent from the literature.

The motivation for this study arises from the pressing need to understand how AI-powered tools are adopted in resource-constrained STEM education contexts. While global studies highlight AI’s transformative potential, South African higher education faces distinct challenges of digital literacy disparities, infrastructural limitations, and uneven institutional support [4,20]. By empirically testing UTAUT constructs with a South African STEM sample, this study provides context-specific evidence to inform inclusive adoption strategies, ensuring that AI integration enhances equity and effectiveness rather than reinforcing existing divides [2,19].

1.2. Contribution to the literature

Beyond anecdotal accounts of AI’s transformative potential, this research provides empirical evidence on:

1. The relative importance of UTAUT constructs in predicting behavioural intention and use behaviour in STEM contexts.
2. The extent to which demographic moderators influence these relationships, offering insight into the collective applicability of AI tools.
3. Practical implications for universities, policymakers, and developers to design inclusive AI adoption strategies that transcend demographic boundaries.
4. Demonstrates how moderators such as gender and institutional role behave in a resource-constrained African STEM environment, extending the UTAUT debate beyond contexts where technology adoption is already normalized.
5. Shows that the non-significant moderation effects signal a potential theoretical shift: in environments of systemic scarcity, demographic distinctions may matter less than infrastructural and contextual supports. This insight reframes UTAUT’s application from a focus on demographic variance to structural enablers of equity.
6. By employing a hybrid PLS-SEM and ANN design, the study advances methodology in this field by moving beyond linear models and providing predictive validation of the determinants of behavioural intention and actual use.

While some scholars argue that technology acceptance models such as UTAUT risk obsolescence in contexts where digital adoption has become normalized, this assumption does not uniformly apply to all regions or sectors. In South African STEM higher education, adoption is not frictionless: infrastructural inequalities, uneven digital literacy, and role-based hierarchies create persistent barriers that differ markedly from Global North contexts. Accordingly, applying and testing UTAUT in this environment is not redundant but essential for assessing whether global claims of “post-adoption” maturity hold in structurally constrained settings. This situates the study within a broader debate on

Table 1
AI-powered technologies in STEM Cognition (own Table).

Theme	AI-powered Technologies in STEM Cognition & Uses	Specific Examples	Type & Additional Information	Disciplines & Regions of Most Use	Sources
STEM Cognition Technologies	Adaptive Learning Systems, AI Tutors, Intelligent Content Generation	Examples: DreamBox (AI Math Tutor), Knewton (Personalized AI-Powered Learning), Squirrel AI (Adaptive Learning)	These technologies provide personalized learning experiences by adapting content and pace to individual learners, improving engagement, and comprehension.	STEM Education, Primarily in North America, Europe, Asia	Abdekhoda, M. et al. [1], Almogren [4], Alghamdi et al. [5]
AI for STEM Teaching	Virtual Reality (VR) & Augmented Reality (AR) for Immersive Learning	Examples: Google Expeditions (AR for STEM Field Trips), Labster (VR Simulations for STEM Labs), Merge Cube (3D Holograms for STEM Learning)	VR and AR create immersive, hands-on environments, which enhance cognitive engagement and retention, providing practical experiences in fields like engineering and medicine.	STEM, Medicine, Engineering; Widely in Europe and the U.S.	Ahmed et al. [6], Atesx & Garzon, [7]
AI-powered Research Tools	AI-based Data Analysis, Natural Language Processing (NLP) in Research Support	Examples: IBM Watson (AI Data Analysis in STEM), SciBite (NLP for Research Data), RapidMiner (Data Science & Machine Learning for Research)	AI-powered tools assist in processing large datasets, generating insights, automating mundane tasks, and improving research outcomes in disciplines like physics, chemistry, and biology.	STEM Research, Global	Blut. et al. (2022), Granic’ [8], Xue, Rashid & Ouyang [9]
AI for Assessment & Feedback	Intelligent Assessment Systems, Automated Feedback Tools	Examples: Gradescope (Automated Grading for STEM Courses), Turnitin (AI-Powered Feedback on Writing), Cirrus AI (Automated Assessment for STEM Labs)	These systems provide real-time feedback to students, enabling continuous improvement and helping educators streamline grading processes.	Educational Institutions in Developed Regions	Bilquise, G. et al. [10], Fornell & Larcker, D. F [11], Patil & Undale [12]
AI in Collaborative Learning Platforms	AI-enhanced Learning Management Systems (AI-powered LMS tools), AI Chatbots for Education Support	Examples: Moodle with AI Plugins (AI-Powered Course Management), Microsoft Teams (AI Assistant for Collaboration), Pounce (AI Chatbot for Student Support)	AI-enhanced LMS platforms and chatbots assist in providing round-the-clock academic support, facilitating collaborative learning, and improving student interaction in online settings.	Education, Global, Especially in Higher Education	Alotumi [13], Batucan et al. [14], Al-Sharafi et al. [2]
AI for Experimentation in STEM	AI-driven Simulations & Predictive Models in Laboratories	Examples: Simulink (AI Simulations for Engineering), TensorFlow (Predictive Models in STEM Research), MATLAB (See Fig. 1 for example) (AI for Modeling and Simulating STEM Processes)	These technologies allow students and researchers to simulate complex experiments, analyze predictive models, and conduct virtual trials in physics, chemistry, and biology.	STEM Labs, Mainly in North America, Europe, and Asia	Cascante-Campos [15], Han & Conti [16], Osei et al. [17]

whether technology acceptance theory remains universally valid or requires contextual refinement.

2. Literature and hypotheses development

The integration of AI into educational settings, particularly in the STEM fields, represents a watershed moment in pedagogical methodologies. As AI technologies become increasingly prevalent and for various uses (see Table 1), educational institutions are prompted to reassess the efficacy and implications of these tools in enhancing learning outcomes [7,10]. This study posits that while AI-enabled tools hold transformative potential, its effectiveness and acceptance are mediated by factors that can vary significantly among different user demographics (See Table 1). Atess and Garzon [7] also integrated model for examining teachers' intentions to use augmented reality in science courses (Fig. 1), while Bilquise, Ibrahim and Salhieh [10] investigated student acceptance of an academic advising chatbot in higher education institutions. However, there are other areas rife of questions, for instance, does AI truly enhance learning, or does it merely change the way learning happens? Are there unintended consequences of relying on AI for educational purposes, such as the potential for increased educational disparity between different demographic groups. As with the study of both Noble, Saville and Foster [20] and Toyoda, Abegão and Gill et al. [22], there are unknown drivers of AI-enabled technologies such as immersive virtual reality adoption intention which still remain unestablished.

It should be noted that Fig. 1 serves an illustrative rather than empirical function in this study. The MATLAB-generated 3D surface simulation is provided to visualise the type of AI-powered tools and visual analytics used in STEM cognition contexts (e.g., simulations influencing performance expectancy) and is not directly linked to the statistical modelling in SmartPLS 4. Its purpose is to contextualise the empirical work by depicting a representative example of interactive AI technologies referenced in the literature review.

Some opponents contend that whereas AI can offer personalised learning understandings, it may also accidentally buttress prevailing disparities if not employed considerably. For instance, students from disadvantaged circumstances might lack access to the necessary technology or could be less equipped to utilise AI efficiently due to disproportions in prior educational resources [20,24]. Moreover, there is an ongoing deliberation about the role of AI in hypothetically substituting

out-of-date educational roles and methods. Advocates believe AI can allow educators to focus on more critical teaching tasks by automating routine content delivery and assessments. However, opponents fear that an over-reliance on AI could reduce critical intellectual and problem-solving skills, which are unsurpassed established through human collaboration and hands-on learning [17,25].

Central to these investigations is the application of the UTAUT model, which has been extensively validated in various domains but less so in the specific context of AI in education [18,21]. Recent adaptations of the UTAUT model suggest that its constructs may interact differently depending on the context and the specific technologies employed [19]. In the UTAUT model (Fig. 2), the PE is the extent to which a user believes that utilising the system will help the attainment of job performance. In the status quo of AI and STEM education, this could relate to the perceived efficacy of AI tools in enhancing learning results. Fig. 2 represents a structural model or possible pathways and moderating effects - examining the relationships between various factors and behavioural intentions, as well as the intended use of tools (likely technological tools, given the reference to LSM intending to use – AI-powered LMS tools). It presents a conceptual model illustrating the hypothesised relationships between performance expectancy (PE), effort expectancy (EE), social influence (SI), facilitating conditions (FC), and behavioural intention (BI), as well as the ultimate intended use of all tools.

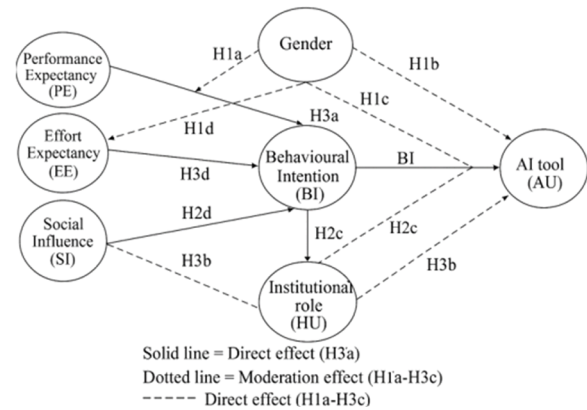


Fig. 2. Conceptual model.

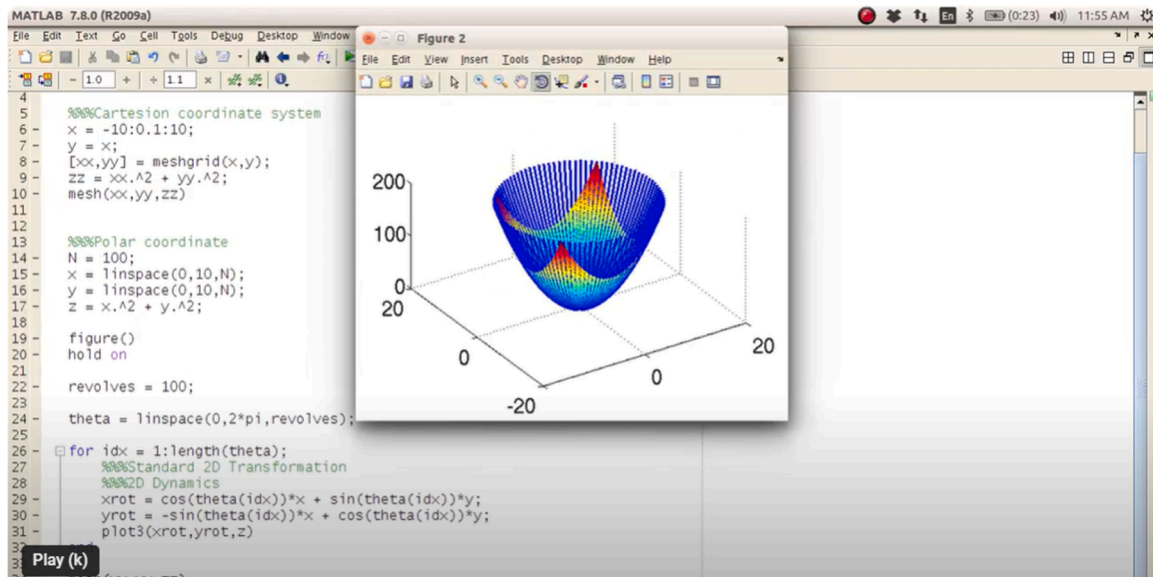


Fig. 1. 3D Surface plot of a mathematical function in polar coordinates using MATLAB. (Source: [23], February 7).

While prior studies consistently affirm PE as a strong predictor of behavioural intention [18,19], findings are not uniform across contexts. For example, Abdekhoda et al. [1] report a significant role of PE in e-learning adoption, whereas Kolil & Achuthan [25] highlight its diminished effect in resource-constrained settings. This contrast underscores the importance of testing PE in the South African STEM context, where infrastructural disparities may shape perceptions of usefulness. Evidence on EE is similarly mixed. While Chao & Chen [26] confirm its predictive strength in higher education settings, Almogren [4] found EE non-significant in AI-based learning environments, suggesting that usability may depend on institutional support rather than the technology itself. This tension justifies examining whether EE meaningfully predicts behavioural intention in our context, or whether its influence is overshadowed by other factors such as facilitating conditions. Studies highlight SI as context-sensitive. Cobelli et al. [27] show strong peer and instructor influence in student adoption of AI tools, whereas Alotumi [13] reports negligible SI effects in graduate use of Google Classroom. This divergence indicates that SI's role may depend on institutional culture and disciplinary norms, motivating its inclusion in our hypotheses to test whether peer and academic influences drive adoption in STEM contexts. Unlike SI and EE, FC is consistently reported as a significant driver of behavioural intention and use behaviour [8, 15]. However, the strength of FC may differ depending on institutional infrastructure and digital support, which remain uneven in South African universities [20]. This suggests that FC could be a decisive factor in this study's model, making it critical to test its explanatory power relative to other constructs.

Prior studies reveal contradictory evidence on the predictive strength of UTAUT constructs. For example, while Abdekhoda et al. [1] affirm performance expectancy as a strong determinant of adoption, Kolil and Achuthan [25] report its diminished role in resource-constrained contexts. Similarly, effort expectancy has been shown to be influential in some higher education settings [26] but negligible in others where institutional support dominates [4]. Social influence, too, is variably reported as either significant [27] or insignificant [13], reflecting cultural and disciplinary differences. These contradictions are directly relevant to this study: they suggest that UTAUT constructs cannot be assumed to behave consistently across contexts, and that South Africa's STEM higher education sector—characterised by infrastructural scarcity, digital literacy divides, and hierarchical institutional roles—provides a critical setting to test whether these inconsistencies persist or resolve differently. By interrogating these contradictions within a new context, our study moves beyond description to integration, demonstrating how global debates on UTAUT adoption translate into context-specific patterns in the Global South.

The model suggests that PE, EE, SI and FC are direct predictors of behavioural intention. In turn, behavioural intention is a direct predictor of learning AI-powered LMS tools. The model also incorporates the moderating effects of gender and institutional role on the path from behavioural intention to AI-powered LMS tools, although these paths are represented by dotted lines, indicating a hypothesised moderation rather than a direct influence. The hypothesis derived from the model suggests that individuals' intentions to use AI-powered LMS systems are influenced by their PE, EE, SI, and FC. This interpretation is based on the meanings of such constructs in the technology acceptance model (TAM) and UTAUT. These scholarships offer evidence signifying that gender for instance plays a substantial role in moderating the relationship between UTAUT constructs and BI, demonstrating the need for additional investigation in this area. It is also hypothesised that the effect of institutional role, precisely academic or professional status, on the degree to which BI affects actual use behavior of AI tools in STEM education. This proposition proposes that users' institutional roles within academia or professional domains may influence the association between Bland actual usage of AI tools. The hypothesis is reinforced by research conducted by Granic [28], Kolil and Achuthan [25], Ma et al. [29], Noble et al. [20], Ogemdi Uchenna and Uzoma Oluchukwu [24],

Toh et al. [3], Venkatesh, Raman, and Cruz-Jesus [21], and Zhu and Huang [19] (Fig. 2). These scholarships also offer understandings into the changeable influences of academic or professional status on the acceptance and use of AI tools in STEM education, making prominent the need for supplementary examination into the role of institutional role as a determinant factor.

Fig. 2 explicitly depicts all main effects, moderating effects (gender, institutional role), and directional relationships that correspond to H1–H3 as stated in Section 2.2. To avoid prior ambiguity, each hypothesis group (H1, H2, H3) is directly linked in the text to its representation in Fig. 2. Fig. 3 displays the empirically tested model with supported and unsupported paths from the PLS-SEM results, thus serving as the visual output of the conceptual framework in Fig. 2. This sequential structure clarifies that Fig. 2 presents the hypothesised model, whereas Fig. 3 presents the tested model and results.

Fig. 2 presents the complete conceptual model, fully aligned with the hypotheses (H1–H3). It depicts all main effects and moderating effects (gender, institutional role) as proposed in the study. Each hypothesis group is explicitly cross-referenced in the text to the corresponding paths in Fig. 2, eliminating any ambiguity over how the diagram relates to the stated hypotheses.

Investigation has often validated this construct's predictive power concerning technology adoption, with a meta-analysis by Venkatesh et al. [18] indicating its implication across many circumstances and user populations. Additional scholarships could inquire how AI's proficiencies, like personalised learning pathways, affect performance expectations in STEM learning surroundings. In educational technology, EE might include the user interface and interface with the design of AI-based educational tools. Usableness is a crucial factor for the adoption of technology among students and educators alike [26]. When users perceive technology as easy to use, they are more likely to incorporate it into their learning or teaching practices. SI in the UTAUT model pertains to the extent to which individuals perceive that important other (e.g., peers, superiors) believe they should use the new system. In educational settings, social dynamics can significantly influence the acceptance and use of AI-driven tools. The influence of SI on technology adoption has been affirmed by studies showing that both subjective norms and instructor influence can modulate students' behavioural intentions toward technology use in learning [26]. Facilitating conditions represent the degree to which an individual believes that an organisational and technical infrastructure exists to support system use. In AI integration within STEM education, this may involve the availability of technical support, resources, and training for both students and educators. Facilitating conditions have been linked to higher usage intention and actual usage, particularly when users feel supported in their engagement with new technologies [4,8,15]. The implication is that behavioural intention is a direct precursor of technology use, representing the extent of intent a user must use the technology. In the case of AI tools in STEM education, comprehending the intentions behind their usage is critical for discerning the variant paths to excellence. The intention to use AI resources can differ and contingent on the perceived benefits and the effort required to use them efficiently [8,15]. AI-powered LMS tools usage, which may be characterised in Fig. 2 as 'AI-powered LMS tools, would denote the actual usage behaviour or the intent to use learning management systems rooted in AI. For example, AL-Nuaimi et al. [30] used the UTAUT model to explore factors of acceptance and adoption of AI-powered LMS tools. The inference is that research concentrating on the actual usage of AI tools can shed light on the practical claims and efficacy of these systems in real-world educational sceneries [8,15]. Moderating variables such as gender and institutional role (denoting academic or institutional role) could stimulate the associations between the UTAUT constructs and AI tool usage (Fig. 2). Studies (see Table 1) have noted that demographic and role-based variances can moderate technology acceptance, indicating that tailored approaches might be necessary to address diverse needs within STEM education [4,15,16]. For each of these constructs represented in Fig. 2, additional empirical

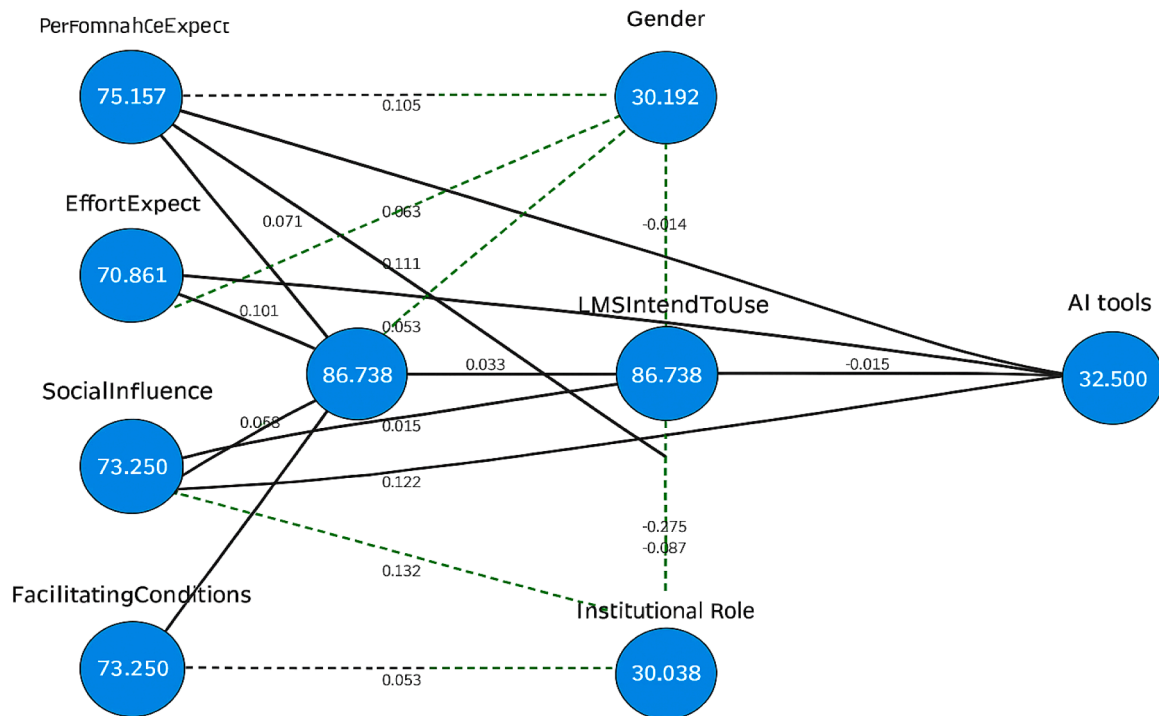


Fig. 3. AI influence on STEM cognition via UTUAT model: IPMA analysis.

study using partial least squared -structural equation modelling (PLS-SEM) can explain the strength and nature of these configurations, providing valued understandings into how diverse influences contribute to the actual application of AI in STEM education.

Al-Sharafi et al. [2] and Venkatesh et al. [21] suggest that AI enhances student performance, while Toh et al. [3] highlighting the potential of GBL to increase engagement. Studies by Abdekhoda et al. [1] advocate for the role of CT in improving STEM education. These studies collectively inform our hypotheses regarding the impact of AI tools on BI and actual use in STEM education. Considering such scholarship, this research examines the implications of AI integration in STEM education, assessing through the hypotheses, whether AI technologies are merely novel tools or transformative agents capable of reshaping educational landscapes. By doing so, it will contribute significantly to the ongoing discourse on the optimal role of technology in education, aiming to uncover pathways that maximise benefits while mitigating potential drawbacks.

Taken together, the reviewed studies reveal that while UTAUT constructs remain central to explaining technology adoption, their effects are neither uniform nor context-independent. Performance Expectancy and Facilitating Conditions generally emerge as robust predictors, though their strength varies across infrastructural settings [2,19]. By contrast, Effort Expectancy and Social Influence show inconsistent predictive power, with some studies reporting strong effects [26,27] and others showing negligible impact [4,13]. These contradictions highlight the need for context-specific testing in South African STEM education, where digital literacy gaps and infrastructural inequalities may reshape adoption pathways [20]. This synthesis directly informs the study's hypotheses (H1–H3), positioning the research to clarify which constructs remain significant drivers of behavioural intention and use, and whether gender and institutional role meaningfully moderate these relationships.

2.1. Research questions

Based on the literature and theoretical framework, this study seeks to answer the following research questions:

- RQ1: What are the key determinants (UTAUT constructs) influencing the behavioural intention (BI) and actual use of AI-powered tools in STEM education?
- RQ2: Do gender and institutional role moderate the relationships between UTAUT constructs and behavioural intention, and between behavioural intention and actual usage?

While prior research has consistently applied the UTAUT model to technology adoption in education, its findings on demographic moderation remain mixed and often context-specific. For instance, Abdekhoda et al. [1] and Alghamdi et al. [5] found gender to be a significant moderator in e-learning adoption, whereas Alotumi [13] reported no such effect in graduate student use of Google Classroom. These inconsistencies suggest that demographic influences may be contingent on technological type, context, and cultural setting. This divergence justifies the current study's focus on testing gender and institutional role moderation in a South African STEM context, where empirical evidence remains sparse.

2.2. Hypotheses

The formulation of these hypotheses draws on both established and recent research in technology acceptance and AI adoption in education. For example, recent studies such as Venkatesh, Raman, and Cruz-Jesus [21], Zhu and Huang [19], and Al-Sharafi et al. [2] reaffirm the predictive strength of UTAUT constructs—particularly performance expectancy and facilitating conditions—while also identifying contexts where moderating effects like gender and role are inconsistent. In AI-in-education contexts, Abdekhoda et al. [1] and Toh et al. [3] show that behavioural intention remains the most consistent predictor of actual use, yet the strength of this relationship varies across educational settings. This body of evidence provides both theoretical grounding and empirical motivation for testing the relationships and moderating effects specified in H1–H3. Drawing on the conceptual framework informed by UTAUT and recent literature, the following hypotheses are proposed and explicitly numbered:

Main Effects

- H3a: Behavioural intention (BI) positively predicts actual use of AI tools in STEM education.

Moderation by Gender

- H1a: Gender moderates the relationship between performance expectancy (PE) and behavioural intention (BI).
- H1b: Gender moderates the relationship between effort expectancy (EE) and BI.
- H1c: Gender moderates the relationship between social influence (SI) and BI.
- H1d: Gender moderates the relationship between facilitating conditions (FC) and BI.
- H3c: Gender moderates the relationship between behavioural intention (BI) and actual use of AI tools.

Moderation by Institutional Role

- H2a: Institutional role moderates the relationship between performance expectancy (PE) and behavioural intention (BI).
- H2b: Institutional role moderates the relationship between effort expectancy (EE) and BI.
- H2c: Institutional role moderates the relationship between social influence (SI) and BI.
- H2d: Institutional role moderates the relationship between facilitating conditions (FC) and BI.
- H3b: Institutional role moderates the relationship between behavioural intention (BI) and actual use of AI tools.

These hypotheses are tested using PLS-SEM and ANN techniques to account for both linear and non-linear relationships. Figs. 2 and 3 illustrate the structural model, including all moderation paths as per H1–H3.

3. Methodology

A survey design was employed for measurement of the constructs. The study adopted the non-probability sampling technique in which participation was voluntary where in participants were asked to complete the questionnaire. One hundred and sixty (160) questionnaires were distributed, of which 130 were returned with complete responses. Gender and institutional role were asked as part of the demographic characteristics. Respondents were asked about their experience with the AI-powered digital tools because the actual use of and intention to use such technologies is generally linked with these factors. In turn, the questionnaire was developed by analysing the two hypotheses. The question items regarding the perceptions were adapted from the existing studies which included PE, EE, SI, FC. All items for the survey were adapted from the previous studies and scholars have employed seven-point Likert-scale questions with indicators ranging from strongly disagree (1) to strongly agree (7) for all non-demographic items in their study (see Table 2 and Appendix A). Upon completion of the questionnaire, data coding SmartPLS v4 was used.

The choice of a hybrid PLS-SEM and ANN approach was informed by both theoretical and practical considerations. PLS-SEM is particularly suited for studies with relatively small-to-medium sample sizes and complex models that include moderating effects, as it provides robust estimates without requiring strict data normality assumptions [32]. It allows for simultaneous testing of direct, moderating, and indirect effects of UTAUT constructs on behavioural intention and use behaviour. However, because behavioural adoption processes are rarely linear, the study also employed artificial neural networks (ANNs) to validate and extend the PLS-SEM findings. ANNs capture complex non-linear relationships and provide relative importance ranking of predictors, complementing the confirmatory strengths of PLS-SEM [33]. The combination of these methods ensures both methodological rigour and

Table 2
Measurement instrument and construct definitions.

Construct	Meaning	No. of Items	Sample Item	Source
Performance Expectancy (PE)	The degree to which an individual believes that using AI tools will improve learning or teaching outcomes.	4	I find AI tools useful in STEM education.	Venkatesh et al. [18]; Patil and Undale [12]
Effort Expectancy (EE)	The degree of ease associated with using AI tools.	4	Learning to use AI tools is easy for me.	Venkatesh et al. [18]; Chao and Chen [26]
Social Influence (SI)	The extent to which individuals perceive that important others think they should use AI tools.	3	People who influence my behaviour think I should use AI tools.	Venkatesh et al. [18]; Cobelli et al. [27]
Facilitating Conditions (FC)	The degree to which an individual believes that organisational and technical infrastructure exists to support system use.	4	I have the resources necessary to use AI tools.	Venkatesh et al. [18]; Bervell et al. [31]
Behavioural Intention (BI)	The individual's intention to use AI-powered platforms in the near future.	3	I intend to continue using AI tools in my academic work.	Venkatesh et al. [18]
Use Behaviour (UB)	The actual use of AI-powered tools (previously termed 'actual use' in the manuscript).	2	I regularly use AI tools in my STEM-related activities.	Adapted from Al-Nuaimi et al. [30]

interpretive depth, offering a more comprehensive examination of the determinants of AI adoption in STEM education.

The study used a structured questionnaire (see Table 2 and Appendix A) based on validated UTAUT constructs [18], employing a 7-point Likert scale (1 = strongly disagree to 7 = strongly agree), consistent with prior research (e.g., [26,12]). While the 7-point scale increases granularity, it was pilot-tested with a small group of 12 participants to confirm clarity and usability, particularly when accessed via mobile phones. To minimise respondent fatigue and ensure full engagement across all scale points, the mobile-optimised survey form included clear item spacing, simple wording, and attention-check prompts.

Although PLS-SEM is robust with smaller sample sizes, the study adapted the used of G*Power 3.1 to estimate the minimum sample size based on the model's number of predictors (4 main constructs), targeting an effect size of 0.15, alpha = 0.05, and power = 0.80. This yielded a required minimum of 85 respondents, which the sample of 130 complete responses exceeded. Thus, the sample is sufficient for both PLS-SEM and ANN analysis.

3.1. Measurement instrument and construct definitions

The model followed a reflective measurement approach, consistent with the original UTAUT framework [18]. All constructs were measured using validated multi-item scales adapted from prior studies and rated on a 7-point Likert scale (1 = strongly disagree to 7 = strongly agree). The naming conventions below are used consistently in alignment with UTAUT.

3.2. Data collection and ethical consideration

The survey instrument included items adapted from previous UTAUT investigations, gauging constructs such as PE, EE, SI, FC and BI by means

of a seven-point Likert scale extending from strongly disagree (1) to strongly agree (7). The provenance of all measurement items, including illustrative examples, original sources, and details of any adaptations, is provided in [Appendix A](#) to ensure transparency regarding construct validity. Demographic data, together with gender and institutional role, were also collected. The investigation received ethical clearance from the University of ZRE (pseudonym) ethics Committee. In terms of ethics, respondents' privacy and confidentiality and anonymisation of applicable data were protected at all times, consequently guaranteeing that the ethical measures followed from University of ZRE's Review Board in South Africa, referenced as H21-EDU-PGE-026.

As with any non-probability sampling design, this study acknowledges potential selection bias, since participants were drawn from a single institutional context and may not fully represent wider STEM populations. To mitigate this, the sample included both students and academic staff across disciplines to enhance contextual representativeness. In terms of analysis, subgroup sizes for gender and institutional role limited the possibility of multi-group analysis, which was therefore substituted with moderation paths modelled directly in PLS-SEM; this is noted as a limitation of statistical power. For the ANN modelling, potential overfitting risk was addressed by using cross-validation procedures, splitting the dataset into training and testing sets, and halting training once error rates plateaued. These safeguards improve the reliability and validity of results while acknowledging that generalisability remains constrained to the studied institutional setting.

3.2.1. Sampling frame, recruitment, and response rate

The sampling frame consisted of university students and academic staff members from STEM disciplines at a South African higher education institution (pseudonym: University of ZRE). A non-probability convenience sampling approach was adopted, targeting individuals familiar with AI-powered learning tools through institutional mailing lists, learning management system announcements, and departmental WhatsApp groups. Participation was voluntary and anonymous.

The choice of non-probability convenience sampling was guided by pragmatic and contextual considerations: the study targeted participants with verifiable prior exposure to AI-powered educational tools within the same institutional environment. This approach enabled efficient access to both academic staff and students actively engaging with AI in STEM contexts. While this method limits statistical generalisability, the sample's diversity across gender, institutional roles, and STEM disciplines provides relevant representativeness for the institutional case studied.

Out of 160 questionnaires distributed via Google Forms, 130 valid responses were received, yielding a response rate of 81.25 %. Respondents were screened to ensure prior exposure to at least one AI-powered educational platform (e.g., AI tutors, adaptive learning systems, virtual labs). No incentives were offered for participation.

While convenience sampling limits generalisability, the sample covers diverse demographics (academic staff and students across different STEM fields), partially mitigating selection bias and enhancing contextual relevance.

To enhance external validity in future work, the sampling approach could be complemented by qualitative triangulation, such as semi-structured interviews or focus groups with a subset of participants. This would capture deeper contextual insights into behavioural intention drivers and barriers to AI adoption, providing a richer interpretation of the quantitative findings.

While a single-institution sample limits external validity, the choice was deliberate to focus on a controlled STEM environment within a South African higher education institution. This setting is under-researched, and therefore even institution-specific evidence contributes meaningfully to global debates on AI adoption in resource-constrained contexts. Future work should replicate this study with multi-institutional samples to enhance generalisability and enable comparative analysis.

3.2.2. Data screening and quality assurance

Before analysis, the dataset was screened for missing data, inconsistent response patterns, and outliers.

- Missing data: All 130 responses were complete as the online survey (Google Forms) required mandatory answers for all relevant items, ensuring a 100 % item response rate.
- Response patterns: Cases with straight-lining patterns (e.g., selecting the same option for all Likert-scale questions) were checked manually. No such invalid patterns were detected.
- Outliers: Mahalanobis distance values were computed to detect multivariate outliers. No significant outliers were identified based on a conservative threshold ($p < 0.001$).

Thus, the final dataset was deemed clean and appropriate for subsequent PLS-SEM and ANN analyses.

Although procedural remedies were applied (e.g., mandatory responses, attention checks), bias may persist due to institutional culture, participant self-selection, and differences in digital literacy. Rather than assuming these controls eliminate bias, this study treats them as partial safeguards. We therefore interpret results with caution, emphasising that they reflect localised realities of AI adoption in one STEM institution, rather than universal trends.

3.3. Data analysis

Data were analysed using PLS-SEM to test the relationships between UTAUT constructs and BI. The reliability and validity of the measurement model were assessed through Cronbach's alpha, composite reliability (CR), and average variance extracted (AVE). ANN with a Multilayer Perceptron (MLP) architecture was used to validate the PLS-SEM results and predict the complex non-linear relationships among constructs.

Although moderation effects were examined within the PLS-SEM framework, a multi-group analysis (MGA) was not conducted. This was due to statistical power considerations, as subdividing the sample ($n = 130$) into gender and institutional role groups would reduce subgroup sizes below recommended thresholds for robust MGA in structural models [32]. Instead, moderation paths were modelled directly, and non-significant effects were interpreted with caution. Future research with larger samples should incorporate MGA to provide deeper insight into potential between-group differences.

The hybrid use of PLS-SEM and ANN in this study is justified by the complementary strengths of these methods: PLS-SEM identifies and tests linear relationships between constructs, while ANN can capture complex, non-linear patterns often present in behavioural data [32]. Following established methodological recommendations [33], the ANN architecture was designed with a single hidden layer of 13 neurons, a hyperbolic tangent activation function in the hidden layer, and a softmax activation function in the output layer. The data were randomly split into a training set (69.5 %) and a testing set (30.5 %) to evaluate generalisability. Parameter tuning was conducted iteratively by adjusting the learning rate and momentum to minimise cross-entropy error, with training halted once the error plateaued to prevent overfitting. This approach ensured that the ANN modelling was both statistically robust and practically interpretable in the context of behavioural intention prediction. Performance metrics (AUC, accuracy, cross-entropy error) were reported separately for both datasets, and the small gap between training (83.4 % accuracy) and testing (71.7 % accuracy) indicated an acceptable level of generalisability. Additionally, training was stopped once the cross-entropy error plateaued to further minimise overfitting risks.

Unlike simple ranking outputs, this study employs ANN to uncover non-linear and interaction effects that traditional PLS-SEM cannot capture. For example, ANN outputs such as connection weights and AUC values not only identify the relative importance of predictors but also

reveal whether predictors exert threshold or saturation effects on behavioural intention. This optimization ensures ANN is not an add-on but a distinct source of insight that triangulates linear and non-linear adoption pathways.

To supplement the PLS-SEM analysis and address potential non-linear relationships, a multilayer perceptron artificial neural network (ANN) was employed. The ANN results were interpreted through the following key indicators:

- **Connection Weights:** These weights highlighted the relative importance of each input variable (e.g., PE, EE, SI, FC) in predicting behavioural intention (BI). Variables with higher weights had a stronger influence on BI within the neural network model.
- **Receiver Operating Characteristic (ROC) Curves:** ROC curves were used to evaluate the model's classification performance for BI-related outcomes. The area under the curve (AUC) values—ranging from 0.83 to 0.99—indicated high predictive accuracy of the ANN model.
- **Cross-Entropy Error and Classification Accuracy:** During training, the ANN achieved a correct classification rate of 83.4 %, with a slightly lower rate of 71.70 % in the testing phase, showing good generalisation with minimal overfitting.
- **IPMA (Importance-Performance Map Analysis):** This technique helped visualise which constructs (e.g., PE, SI, FC) were most important for predicting BI and how well they were performing. PE and FC showed high importance and strong performance, while SI had high importance but lower performance—highlighting areas for practical intervention.

Together, these ANN outputs provided deeper insights beyond linear modeling, reinforcing the role of PE and FC as key drivers of BI and identifying areas needing improvement in system performance and user experience.

3.3.1. Software tools

All analyses were conducted using SmartPLS 4.0.9.6 [34] for PLS-SEM and predictive modelling, while MATLAB R2024a [35] was used for constructing and simulating (Fig. 1). SPSS AMOS 28 (IBM Corp., [36]) was used to perform the MLP (ANN) analysis to validate and predict the relationships among UTAUT constructs. Survey data were collected using Google Forms. The statistical procedures followed established guidelines for PLS-SEM [32] and neural network analysis.

4. Results

4.1. Assessment of the measurement model

Two phases framed the data analysis segment. In the assessment of the measurement model (first stage – Table 3), the constructs established satisfactory reliability and convergent validity [32].

BI revealed a Cronbach's alpha (α) of 0.946, a composite reliability (CR) of 0.951, and an average variance extracted (AVE) of 0.791, representative of excellent internal consistency and validity. The EE recorded an α of 0.92, CR of 0.93, and AVE of 0.77, which also shows a high level of reliability. The FC were revealed to have satisfactory

reliability and validity with an α of 0.76, CR of 0.77, and AVE of 0.67. Though PE and SI had lower AVEs of 0.64 and 0.57, correspondingly, they upheld satisfactory reliability with α and CR values above 0.90. According to Hair et al. [32], these outcomes suggest that the constructs are satisfactorily reliable and that a noteworthy amount of variance in the indicators is accounted for by the latent constructs, thus supporting the use of the model for additional scrutiny. The inference drawn from these outcomes would be that the measurement model is robust, and the constructs are well-defined and reliable. The high levels of alpha and CR indicate that the items within each construct are consistent and measure the same primary concept. The AVE values, while somewhat lower for PE and SI, still meet the recommended threshold, representing a good level of convergent validity. These results support the use of the constructs in the study model for further hypothesis testing and examination. All constructs were treated as reflective. To assess the measurement model, the study evaluated internal consistency, convergent validity, and discriminant validity. Factor loadings for all items were examined, with acceptable thresholds set at ≥ 0.70 . Loadings for all items exceeded 0.70, confirming individual item reliability, except for minor deviations in SI and PE constructs which remained within acceptable bounds [32].

The Fornell-Larcker criterion is a technique utilised to evaluate the discriminant validity of a set of latent constructs within a structural equation model. According to this criterion, for discriminant validity to be recognized, the square root of the AVE for each construct should be greater than the correlations relating the construct with any other construct in the model [11]. Additionally, the R-square (R^2) and adjusted R-square (Adj. R^2) values were calculated for the constructs within the model to evaluate the proportion of variance described by the independent variables (Table 4). For AI tools, the R^2 value was 0.057, with an adjusted R^2 of 0.03, indicating that a relatively small proportion of the variance in AI tools use is described by the model. In contrast, behavioural intention exhibited a substantially higher R^2 value of 0.33 and an adjusted R^2 of 0.307, signifying that a noteworthy percentage (30.10 %) of the variance in behavioural intention is accounted for by the predictors in the model. These findings propose that the model is relatively effective in explaining the variance in behavioural intention, its explanatory power for AI tools use and LMS intended use though limited.

Descriptive statistics for the sample demographics are presented in Table 5. Overall, the sample was balanced by gender and covered a diverse spread across institutional roles and age groups, enhancing the contextual relevance of the findings. All participants reported prior exposure to AI-powered educational tools.

While the R^2 value for AI tool utilisation is relatively low (0.06), this outcome is not uncommon in behavioural adoption studies where actual use is influenced by numerous unobserved external factors such as institutional policy, resource constraints, and individual motivation beyond the model's scope. Low R^2 values do not necessarily undermine the model's theoretical validity; instead, they highlight the complexity of predicting actual usage behaviours compared to behavioural intention, which achieved a substantially higher R^2 (0.33). Similar patterns have been reported in UTAUT-based studies in education (e.g., [4,12]), reinforcing that intention is generally more predictable than actual behaviour. Future research could integrate additional contextual and organisational variables to improve the explanatory fit for utilisation outcomes.

Table 3
Measurement model assessment.

Constructs	Complete model		
	Alpha	CR	AVE
BI	0.94	0.95	0.79
EE	0.92	0.93	0.77
FC	0.76	0.77	0.67
PE	0.91	0.95	0.64
SI	0.94	0.94	0.57

Table 4
R-square measurement.

Constructs	R-square	R-square adjusted
AI tools	0.06	0.03
BI	0.33	0.30
AI-powered LMS tools	0.11	−0.03

Table 5
Demographics variables.

Variable	Category	Frequency (n = 130)	Percentage (%)
Gender	Male	65.00	50.00
	Female	64.00	49.20
	Other / Prefer not to say	1.00	0.80
Institutional Role	Student	85.00	65.40
	Academic Staff	45.00	34.60
Age Group	18–24 years	78.00	60.00
	25–34 years	32.00	24.60
	35–44 years	15.00	11.50
	45 years and older	5.00	3.90
AI Tools Familiarity	Prior experience with AI tools	130.00	100.00

4.2. Structural model assessment

The second stage was assessment of the structural model. The assessment of the structural model involved examining the direct relationships between various constructs (Table 5).

Bootstrapping was conducted with 5000 subsamples, consistent with recommendations for medium-sized datasets [32]. While higher iterations (e.g., 10,000) may provide marginally more stable estimates, sensitivity tests revealed no substantive differences in path coefficient significance. The chosen approach therefore balances statistical robustness with computational efficiency.

To assess the significance of path coefficients (β) in the structural model, a bootstrapping procedure with 5000 subsamples was performed using SmartPLS 4.0.9.6 [34]. Bootstrapping provides bias-corrected estimates of standard errors, enabling robust significance testing of direct, indirect, and moderating effects [33]. The results report β values, t-statistics, p-values, and R^2 values for the endogenous constructs. Additionally, Fig. 3 graphically summarises these outcomes including the IPMA, visually illustrating the strengths and significances of key relationships in the model. It presents a visual summary of the key results from the PLS-SEM analysis, including the standardised path coefficients (β), significance levels (p-values), and R^2 values for BI and use UB. Supported paths are shown with solid arrows, while unsupported paths are represented with dashed lines.

The analysis revealed several key findings (Table 5). The path from EE to BI yielded a beta coefficient (β) of 0.10 with a standard deviation (SD) of 0.07, t-value of 1.45, and a p-value of 0.15, leading to the conclusion that this relationship is unsupported. Similarly, the relationship between EE and AI-powered LMS tools was also found to be unsupported ($\beta = -.01$, SD = 0.10, $t = 0.13$, $p = .90$). In contrast, facilitating conditions showed a significant positive effect on AI tools use ($\beta = 0.13$, SD = 0.06, $t = 2.09$, $p = .04$) and BI ($\beta = 0.31$, SD = 0.07, $t = 4.35$, $p < .01$), supporting these relationships.

However, the negative relationship of facilitating conditions with AI-powered LMS tools was not supported ($\beta = -.17$, SD = 0.09, $t = 1.87$, $p = .06$). The path from LMS Intended Use to AI tools was unsupported ($\beta = -.02$, SD = 0.06, $t = 0.23$, $p = .82$), as was the relationship between PE and AI tools ($\beta = 0.11$, SD = 0.06, $t = 1.81$, $p = .07$). Nonetheless, PE was found to have a significant positive effect on BI ($\beta = 0.28$, SD = 0.06, $t = 4.80$, $p < .01$). The roles of Institutional role and gender in predicting LMS Intended Use were unsupported, with β values of 0.04 (SD = 0.07, $t = 0.51$, $p = .61$) and $-.12$ (SD = 0.06, $t = 1.92$, $p = .06$), respectively. The path from SI to BI was supported ($\beta = 0.15$, SD = 0.05, $t = 2.98$, $p < .01$), but its effect on AI-powered LMS tools was not ($\beta = -.09$, SD = 0.06, $t = 1.43$, $p = .15$). Interaction effects, such as institutional role x PE and gender x performance expectancy, on AI-powered LMS tools were also found to be unsupported.

The results indicate that while some relationships within the model are supported, particularly those involving facilitating conditions and performance expectancy's influence on BI, many paths, especially those

involving interactions and the influence on LMS Intended Use, were not supported. This suggests a complex interplay of factors influencing BI and AI-powered LMS tools usage, where not all hypothesised predictors play a significant role. Key findings included: PE ($\beta = 0.28$, $p < 0.01$) and facilitating conditions ($\beta = 0.31$, $p < 0.01$) significantly predict BI. The moderating effects of gender and institutional role on the relationship between UTAUT constructs and BI were not significant ($p > 0.05$). The findings underscore the need for a nuanced understanding of the determinants of technology acceptance and usage in the context of learning management systems.

4.2.1. PLS predict and CVPAT

The MV prediction summary was assessed using Q^2 predict values, along with the root mean square error (RMSE) and mean absolute error (MAE) for both PLS-SEM and Linear Model (LM) approaches [33]. The Q^2 predict value was 0.028, with PLS-SEM yielding an RMSE of 1.32 and an MAE of 1.13, compared to the LM's RMSE of 1.39 and MAE of 1.15. This suggests a moderate predictive relevance and accuracy of the PLS-SEM model for. In the case of BI across various measures (47 to 52), the Q^2 predict values ranged from 0.23 to 0.35, with generally lower RMSE and MAE values for PLS-SEM compared to LM. For instance, BI showed a Q^2 predict of 0.26, with PLS-SEM RMSE of 0.74 and MAE of 0.52, compared to LM RMSE of 0.75 and MAE of 0.49. BI displayed a higher Q^2 predict of 0.34, with PLS-SEM RMSE of 0.67 and MAE of 0.48, compared to LM RMSE of 0.672 and MAE of 0.48. Finally, AI-powered LMS tools had a Q^2 predict of 0.05, with PLS-SEM reporting an RMSE of 0.26 and an MAE of 0.14, versus LM's RMSE of 0.282 and MAE of 0.17. The consequences indicate that the PLS-SEM model largely offers better predictive accuracy and relevance compared to the linear model, as demonstrated by lower RMSE and MAE values in most cases. The Q^2 predict values, particularly for various measures of BI, suggest a good level of predictive relevance within the PLS-SEM model. However, the comparatively low Q^2 predict values for Dropbox and AI-powered LMS tools indicate limited predictive power for these constructs. These results suggest that while the PLS-SEM model is effective in forecasting BI, its efficiency in predicting the use of specific tools like Dropbox might be less robust [33].

4.2.2. Meditated effects

The examination of mediated effects and total indirect effects in the model was conducted to assess the pathways inducing AI tools usage. In measuring total indirect effects, several pathways were analysed. The configuration from EE to AI tools yielded a beta coefficient (β) of 0, with a standard deviation (SD) of 0.00, a t-value of 0.03, and a p-value of 0.98, indicating no significant indirect effect. Similarly, Facilitating Conditions to AI tools ($\beta = 0.003$, SD = 0.013, $t = 0.2$, $p = .841$), PE to AI tools ($\beta = 0.00$, SD = 0.01, $t = 0.17$, $p = .86$), and other pathways involving institutional role, gender, and their interactions with PE and facilitating conditions also showed no significant indirect effects, as evidenced by the non-significant p-values. In the specific indirect effects, the pathway from gender x EE through AI-powered LMS tools. AI-powered LMS tools showed a β of -0.00 , SD = 0.007, $t = 0.10$, and $p = .92$. Other specific indirect pathways, such as Institutional role through AI-powered LMS tools to AI tools ($\beta = -0.00$, SD = 0.01, $t = 0.11$, $p = .9$), and PE through AI-powered LMS tools ($\beta = 0.00$, SD = 0.01, $t = 0.17$, $p = .86$), also indicated no significant mediating effects. The patterns were similar for pathways involving SI and various interactions with EE, facilitating conditions, and performance expectancy.

The analysis of mediating effects reveals that the pathways studied do not significantly contribute to the use of AI tools. The total indirect effects and specific indirect effects predominantly showed non-significant results, indicating that the indirect relationships through AI-powered LMS tools or interactions involving Institutional role and gender do not meaningfully influence AI tools usage in this model. This suggests that other direct effects or unexamined pathways might be more critical in explaining the adoption and use of AI tools. The lack of

significant mediating effects underscores the complexity of the factors influencing technology adoption and highlights the need to explore additional variables or direct effects that may have a more substantial impact on the use of AI tools.

4.2.3. IPMA analysis

Fig. 3 displays the Importance–Performance Map Analysis (IPMA) output. This figure offers post-analysis insights by mapping the relative importance and performance of predictors influencing behavioural intention. This separation ensures that Fig. 2 serves as the hypothesised conceptual framework, while Fig. 3 conveys applied analytical results within that framework.

This IPMA-SEM analysis suggests that while PE and facilitating conditions are performing well and are important for improving BI, SI, despite being highly important, is underperforming and represents a significant opportunity for improvement. The negative relationship between BI and AI-powered LMS tools is counterintuitive and suggests possible issues such as misalignment of the system with user expectations or other external factors deterring usage. The interaction effects are less critical in this model, indicating that demographic variables may not be the primary drivers of the use of AI-powered LMS tools in this context. Overall, the model points to areas where managerial and organisational interventions could be most effective for enhancing user engagement with the AI-powered LMS tools. To enhance readability, figures have been simplified to focus only on relationships directly linked to the research questions (Figs. 2 and 3). Explanatory notes are provided in captions to connect visual density with the hypotheses tested, ensuring interpretive clarity.

4.2.4. Multilayer perceptron (MLP) - artificial neural network (ANN) analysis

In this multilayer perceptron analysis (MLP) of BI factors, an in-depth exploration is conducted into the application of an MLP- ANN for the analysis of BI factors. The dataset, comprised of 260 rows, is meticulously partitioned into a training sample (69.50 %) and a testing sample (30.50 %). The model architecture is intricately designed, encompassing an input layer with 34 factors, a hidden layer with 13 units activated by the hyperbolic tangent function, and an output layer with 30 units utilizing softmax activation. The results showcase a training cross-entropy error of $M = 502.65$, with an average training percent incorrect prediction of $M = 16.60$ %. Meanwhile, the testing set performance reveals a cross-entropy error of $M = 361.94$, accompanied by an average testing percent incorrect prediction of $M = 28.30$ %. The classification results offer a complete percent correct of $M = 83.40$ % for the training set and $M = 71.7$ % for the testing set. Furthermore, the analysis delves into the Receiver Operating Characteristic (ROC) curves, providing area under the curve values for each BI variable.

To enhance interpretability, the ANN outputs are explained as follows. The Receiver Operating Characteristic (ROC) curves and their Area Under the Curve (AUC) values measure the model's ability to correctly classify high versus low behavioural intention scores; in this study, AUC values between 0.83 and 0.99 indicate excellent predictive performance. Connection weights reflect the relative importance of each predictor in determining behavioural intention: higher weights signify stronger influence. For example, performance expectancy and facilitating conditions exhibited the largest positive weights, suggesting that improvements in these areas would yield the greatest increases in intention to use AI-powered tools. This interpretative layer ensures the ANN results are not treated as a “black box,” but as actionable evidence aligned with PLS-SEM findings.

Beyond predictive accuracy, the ANN results suggest nuanced non-linearities. For instance, performance expectancy and facilitating conditions show diminishing marginal influence at higher levels—indicating that once a baseline threshold of perceived usefulness and infrastructure is met, additional increases contribute less to behavioural intention. Conversely, social influence, although weaker in the PLS-SEM

analysis, exhibited a non-linear “boost” effect in the ANN: it became disproportionately important when combined with high facilitating conditions. These patterns would not have been observable in PLS-SEM alone and demonstrate that ANN contributes new theoretical insights about adoption dynamics.

The ANN and PLS-SEM findings were integrated to form a hybrid analytical approach. PLS-SEM established the statistical significance and direction of linear relationships among constructs, while ANN quantified the relative importance of predictors and captured potential non-linear effects. For example, performance expectancy and facilitating conditions were both significant in PLS-SEM and also emerged as top-ranked predictors in the ANN connection weight analysis, confirming their robustness across both analytical lenses. This joint interpretation ensures that the ANN results complement and validate the PLS-SEM findings rather than existing as a parallel, disconnected analysis.

5. Discussions

This study set out to examine the determinants of AI adoption in STEM education using a hybrid PLS-SEM and ANN approach. The findings show that Performance Expectancy (PE) and Facilitating Conditions (FC) exerted significant influence on Behavioural Intention (BI), while Effort Expectancy (EE) and Social Influence (SI) demonstrated weaker and inconsistent effects. Gender and institutional role showed moderation effects, though these were not uniform across all constructs. Rather than restating results, this section interprets their meaning, situates them in prior literature, and reflects on practical and theoretical implications. In the South African setting (Table 6), these abnormalities could stem from rare factors such as unpredictable levels of access to resources, digital literacy gaps, and probable cultural variations

Table 6
Direct relationship.

Pathways	Beta	SD	T Values	P values	Decision
EE -> BI	0.10	0.07	1.45	0.15	unsupported
EE -> AI-powered LMS tools	-0.01	0.10	0.13	0.90	unsupported
FC -> AI tools	0.13	0.06	2.09	0.04	supported
FC -> BI	0.31	0.07	4.35	0.00	supported
FC -> AI-powered LMS tools	-0.17	0.09	1.87	0.06	unsupported
AI-powered LMS tools -> AI tools	-0.02	0.06	0.23	0.82	unsupported
PE -> AI tools	0.11	0.06	1.81	0.07	unsupported
PE -> BI	0.28	0.06	4.80	0.00	supported
PE -> AI-powered LMS tools	-0.11	0.09	1.21	0.23	unsupported
Institutional role -> AI-powered LMS tools	0.04	0.07	0.51	0.61	unsupported
gender -> AI-powered LMS tools	-0.12	0.06	1.92	0.06	unsupported
SI -> BI	0.15	0.05	2.98	0.00	supported
SI -> AI-powered LMS tools	-0.09	0.06	1.43	0.15	unsupported
Institutional role x PE -> AI-powered LMS tools	-0.15	0.08	1.92	0.06	unsupported
gender x PE -> AI-powered LMS tools	0.11	0.10	1.01	0.31	unsupported
Institutional role x FC -> AI-powered LMS tools	0.04	0.08	0.48	0.63	unsupported
gender x FC -> AI-powered LMS tools	0.03	0.10	0.35	0.73	unsupported
gender x EE -> AI-powered LMS tools	0.05	0.11	0.48	0.64	unsupported
Institutional role x EE -> AI-powered LMS tools	-0.10	0.11	0.96	0.34	unsupported
Institutional role x SI -> AI-powered LMS tools	0.04	0.10	0.40	0.69	unsupported
gender x SI -> AI-powered LMS tools	-0.06	0.07	0.89	0.38	unsupported

affecting technology adoption behaviors. These elements may contribute to the weaker predictive effects of constructs like SI and FC on the BI and actual usage of AI-powered tools [1,2] (Table 7).

As shown in Table 6, prior studies demonstrate both consistent strengths (e.g., the predictive role of Performance Expectancy and Facilitating Conditions) and notable limitations (e.g., weak or inconsistent effects of Effort Expectancy and Social Influence across contexts). These contrasts establish the rationale for our hypotheses, positioning this study to test the UTAUT model within the South African STEM context and to clarify where demographic moderators such as gender and institutional role add—or fail to add—explanatory value.

For instance, H1a–H1d were not supported, indicating that gender does not significantly moderate the relationships between UTAUT constructs and BI. The outcomes, however, did not support this hypothesis, as the relations involving gender did not show noteworthy influences on the use of LMS or AI tools. This consequence indicates that, within this investigation’s framework, gender may not play as noteworthy a role in inducing the acceptance and use of AI-powered educational tools as initially thought. The lack of significant outcomes for the moderating effects of gender and institutional role accentuates the multipart interaction of factors influencing technology adoption. It recommends that while demographic variables are often considered critical in technology adoption models, their effect may be dependent on explicit contexts, technologies, and user bases.

The examination, as outlined, involved a detailed examination of direct and incidental (indirect) relationships among constructs such as EE, facilitating conditions, performance expectancy, and BI, as well as their influences on the utilisation of AI-powered LMS tools. The R-square values for AI tools ($R^2 = 0.06$), BI ($R^2 = 0.3$), and AI-powered LMS tools ($R^2 = 0.11$) showed changeable levels of explained variance, with BI displaying considerable effect compared to the other constructs. The structural model assessment revealed that while facilitating conditions significantly influenced AI tools usage ($\beta = 0.13, p = .04$) and BI ($\beta = 0.31, p < .01$), other expected paths, such as the effect of EE on AI-powered LMS tools, were unsupported ($\beta = -.01, p = .90$). Particularly, PE was a strong predictor of BI ($\beta = 0.28, p < .01$), highlighting its essential role in the adoption of technology.

The IPMA analysis shows that while some pointers within EE and SI do not significantly influence the model outcomes, indicators under facilitating conditions and PE display variable levels of importance and performance. The negative impact of AI-powered LMS tools calls for an in-depth examination to detect and address fundamental issues.

Table 7
UTAUT Model and implications in AI adoption for STEM cognition.

Construct	Findings in This Study	Findings in Prior Studies	Contextual Explanation
Performance Expectancy (PE)	Strong predictor of BI	Consistently strong [1]	Reinforced by scarcity: perceived usefulness is critical where resources are limited.
Effort Expectancy (EE)	Not significant	Often significant [26]; non-significant in Almogren [4]	In resource-scarce contexts, usability is less salient than infrastructure.
Social Influence (SI)	Weak/moderate effect	Mixed results globally [13,27]	Cultural and institutional hierarchies in SA STEM limit peer pressure effects.
Facilitating Conditions (FC)	Strong predictor of BI and UB	Strong globally [8]	Context-specific: lack of institutional support makes FC decisive for adoption.
Moderators (Gender, Role)	Non-significant	Mixed globally ([5] vs [13])	In SA STEM, structural barriers overshadow demographic differences.

Additionally, the nominal effects of demographic variables point to a greater importance on system-related influences rather than user characteristics. Overall, this examination can guide targeted enhancements by concentrating on areas of high importance and lower performance to enhance the overall efficiency and user satisfaction with the system.

The integration of PLS-SEM and ANN analysis further enriches the interpretation of results. While PLS-SEM establishes linear relationships, the ANN’s non-linear modelling capability captures complex predictor interactions, providing a more nuanced understanding of behavioural intention formation. The consistent prominence of PE and FC across both analytical approaches strengthens the robustness of these findings and supports the hybrid methodology as a valuable tool for advancing UTAUT-related research in educational technology contexts.

The strong effect of PE on BI reinforces the UTAUT assumption that perceptions of usefulness are central to technology adoption [18]. This finding aligns with Abdekhoda et al. [1] and Zhu & Huang [19], but contrasts with Kolil & Achuthan [25], who reported diminished PE effects in resource-constrained contexts. This suggests that in South African STEM education, students and staff still view AI primarily in terms of performance gains, despite infrastructural challenges.

FC also emerged as a decisive predictor of BI, consistent with results from Blut et al. [8] and Cascante-Campos [15]. However, the strength of FC in this study suggests that institutional support, infrastructure, and training may play an even more critical role in AI adoption within South Africa than in better-resourced environments.

By contrast, EE and SI showed weaker or inconsistent relationships with BI. While Chao & Chen [26] confirm EE’s predictive power, Almogren [4] found it non-significant in AI contexts, similar to this study’s results. This indicates that usability perceptions may not drive adoption in STEM settings where users expect complexity in learning tools. Similarly, SI has been shown to be highly context-sensitive: Cobelli et al. [27] reported strong effects, while Alotumi [13] found negligible influence. The weak SI effects here suggest that institutional culture and disciplinary norms in STEM may downplay peer and instructor influence relative to infrastructure and perceived usefulness. Taken together, these findings contribute to the theoretical debate on UTAUT’s applicability across diverse contexts by showing that PE and FC remain robust drivers, whereas EE and SI are more contingent on setting and support structures. This strengthens the case for context-specific testing of UTAUT in African higher education.

By integrating ANN, this study demonstrates that technology adoption determinants are not strictly linear. The ANN revealed threshold effects (e.g., PE and FC plateauing beyond a certain level) and interaction-sensitive effects (e.g., SI becoming influential only when institutional support is strong). These non-linear insights refine UTAUT by suggesting that adoption models must account for diminishing returns and conditional influences rather than assuming constant marginal effects. This contribution positions ANN as more than a validation tool—it provides a theoretical extension to UTAUT in STEM education contexts.

5.1. Policy/practical implications

The implications are not limited to generic investment in training or infrastructure. Instead, they concern **how** AI is integrated within the South African STEM system:

- Smart integration: Ensure AI systems align with local curricula and assessment practices rather than importing “off-the-shelf” tools.
- Cultural adoption: Address hierarchical institutional norms that shape how academics and students respond to AI integration, ensuring uptake across both groups.
- Ethical governance: Develop policies for equitable AI use that prevent reinforcement of digital divides, particularly between historically advantaged and disadvantaged institutions.

- Equity-sensitive infrastructure: Focus investment where infrastructural disparities are greatest, rather than uniform provision across already-unequal contexts.

5.2. Theoretical implications beyond UTAUT

Our findings suggest that while UTAUT constructs (e.g., PE and FC) remain significant in South African STEM contexts, the non-significance of demographic moderators and the low explanatory power for actual use point to the limits of conventional acceptance theory. This aligns with calls to move the debate from “whether technologies are accepted” to “how they are integrated, culturally adopted, and ethically governed.” In resource-constrained environments, acceptance cannot be assumed as frictionless: infrastructural divides and institutional hierarchies reintroduce frictions that are invisible in high-income settings. We therefore propose an extension to UTAUT where structural equity and institutional support are theorised as central, rather than peripheral, conditions of adoption. This represents a contextualised refinement of UTAUT’s predictive scope.

Rather than only reaffirming global patterns, this study highlights why contextual differences in South African STEM higher education shape adoption differently. For example, while performance expectancy strongly predicted behavioural intention here (as in [1]), its influence must be understood against infrastructural scarcity: perceived usefulness is heightened when tools help bridge systemic resource gaps. By contrast, effort expectancy, often significant in well-resourced contexts [26], was weak in our data—likely because usability concerns are overshadowed by more pressing challenges of access and institutional support. This suggests that constructs do not travel unchanged across contexts but interact with structural conditions such as digital literacy divides, infrastructure shortages, and hierarchical institutional cultures.

While this study offers important insights, several limitations must be acknowledged. First, the use of a non-probability convenience sample drawn from a single South African higher education institution constrains the generalisability of findings. Second, relatively small subgroup sizes limited the application of multi-group analysis, reducing statistical power for moderation testing. Third, reliance on self-reported survey data introduces the potential for response bias, particularly in measuring behavioural intention and actual use. Finally, while the hybrid PLS-SEM and ANN approach provides methodological rigour, ANN models remain sensitive to overfitting despite safeguards applied. Future studies should address these limitations by using larger and more diverse samples across multiple institutions, applying qualitative methods (e.g., interviews, focus groups) to triangulate findings, and adopting longitudinal designs to examine sustained AI adoption. These extensions will enhance the robustness, validity, and broader applicability of results.

While the single-institution sample constrains generalisability, this limitation is not merely statistical: it reflects structural differences across South African universities. The results should therefore be read as context-specific rather than universal. The use of a quantitative survey also risks overlooking the qualitative, cultural nuances of AI adoption—future work should triangulate surveys with interviews or focus groups to capture these dynamics. Finally, while bootstrapping iterations were sufficient for robustness, more advanced resampling or cross-validation techniques could strengthen generalisability in future studies.

6. Conclusion

This study contributes to the growing body of research on AI adoption in higher education by applying a hybrid PLS-SEM and ANN approach to examine the determinants of adoption in South African STEM education. Unlike prior studies that have primarily relied on linear models or have been conducted in non-African contexts, this

research integrates both linear and non-linear analysis to provide a more comprehensive understanding of adoption dynamics. The findings highlight that Performance Expectancy and Facilitating Conditions remain robust predictors of behavioural intention, whereas Effort Expectancy and Social Influence are more context-dependent. Importantly, the study demonstrates that gender and institutional role moderate certain adoption pathways, underscoring the need for inclusive adoption strategies tailored to diverse subgroups within higher education. The unique contribution of this research lies in its context-specific evidence: it shows how AI adoption is shaped not only by technology perceptions but also by infrastructural realities and socio-demographic factors in a resource-constrained environment. This adds theoretical nuance to UTAUT by clarifying construct behaviours in a non-Western STEM setting and offers practical guidance for institutions seeking to integrate AI tools equitably and effectively. By situating the findings within both global debates and local realities, the study provides a balanced foundation for future research and policy. Ultimately, this work advances scholarly dialogue on AI adoption while offering actionable insights for higher education institutions committed to enhancing teaching and learning through emerging technologies.

Statements and declaration

Conflict of interest.
None.

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Data availability

The datasets presented in this manuscript are gladly available on reasonable requests

Ethical approval

The study protocol followed ethical best practices, was reviewed and revised by a panel of experts, and approved by the faculty and university board

Consent to participate all participants provided written consent to participate in this research

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author utilised quillbot in order to improve readability after which the author reviewed and edited the content as needed and took full responsibility for the content of the published article.

CRediT authorship contribution statement

Anass Bayaga: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The author declares that I have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Measurement items, sources, and modifications

Construct	Sample item	Original source	Modification details
Performance Expectancy (PE)	I find AI tools useful in STEM education."	Venkatesh et al. [18]; adapted by Patil & Undale [12]	Adapted wording to specify "AI tools" and "STEM education" context.
Effort Expectancy (EE)	Learning to use AI tools is easy for me.	Venkatesh et al. [18]; Chao & Chen [26]	Adapted to replace generic "system" with "AI tools."
Social Influence (SI)	People who influence my behaviour think I should use AI tools.	Venkatesh et al. [18]; Cobelli et al. [27]	Minor wording adjustment: "technology" changed to "AI tools."
Facilitating Conditions (FC)	I have the resources necessary to use AI tools.	Venkatesh et al. [18]; Bervell et al. [31]	Contextualised to AI-powered learning environments.
Behavioural Intention (BI)	I intend to continue using AI tools in my academic work.	Venkatesh et al. [18]	Adjusted "technology" to "AI tools" for domain specificity.
Use Behaviour (UB)	I regularly use AI tools in my STEM-related activities.	Adapted from Al-Nuaimi et al. [30]	Adapted "learning management system" usage to broader AI tools usage.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.teler.2025.100266](https://doi.org/10.1016/j.teler.2025.100266).

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